

**SAME RESIN. SAME COLUMN.
SAME PACKING VELOCITY.
DIFFERENT PACKING OUTCOME.**

**How column packing protocol
influences IEC column efficiency.**

**A Case study
Protpure Tech Pvt Ltd**



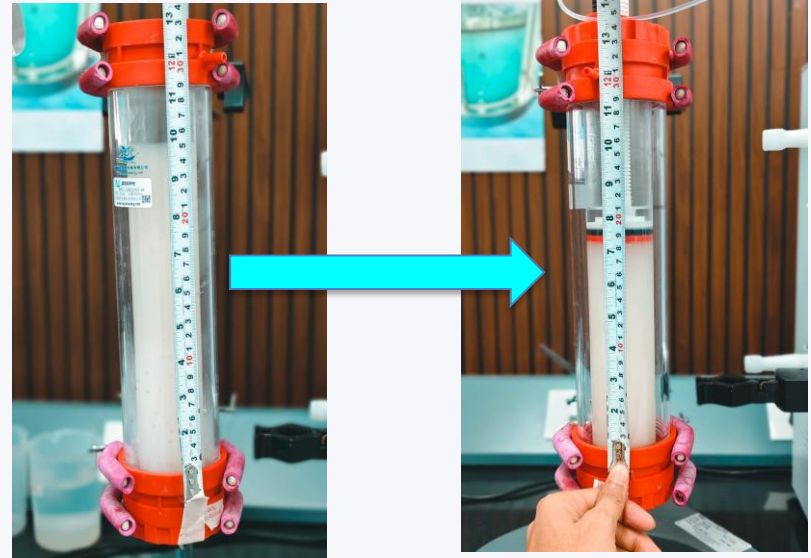
Why Column Packing Matters

In chromatography, reproducibility does not start with sample loading.

It starts with column packing.

Even when using:

- ☐ Same resin
- ☐ Same column
- ☐ Same buffer system
- ☐ Similar packing velocities



Small variations in packing workflow and bed consolidation can significantly influence:

- ✓ Column efficiency
- ✓ Peak symmetry
- ✓ Process reproducibility
- ✓ Scale-up consistency

Experimental Conditions

Column Parameters

Column: 50/300 mm

Column Diameter: 5 cm

Resin Volume: 353 mL

Slurry Height: 25 cm

Packing Buffer

20 mM Tris 50 mM NaCl, pH 7.5

Column Evaluation

Injection: 2 mL of 1% acetone

UV monitoring at 280 nm

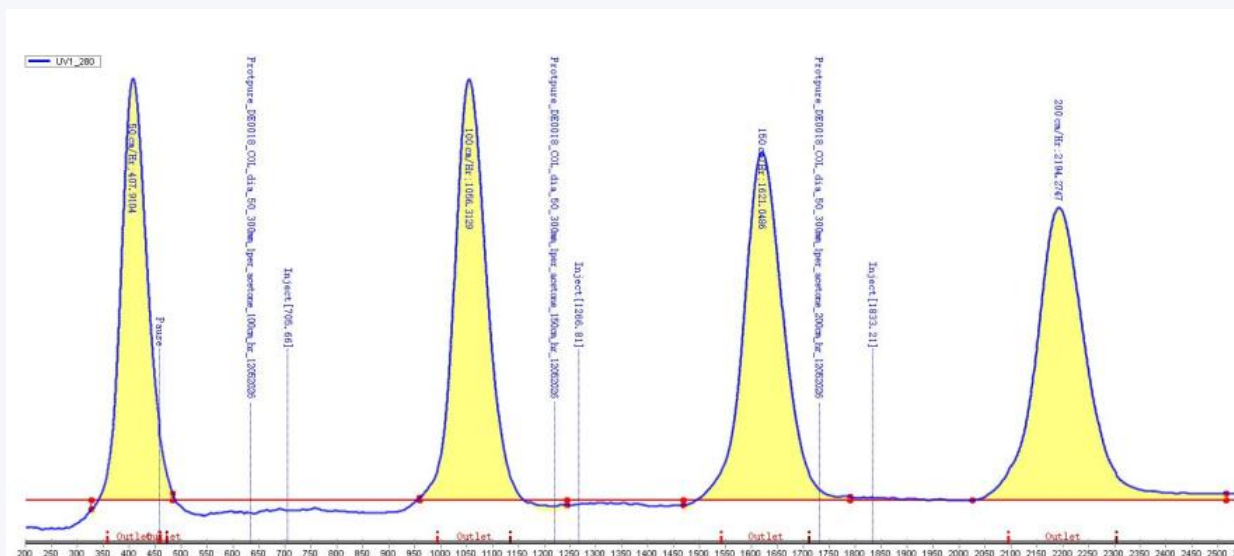


Study Objective:

“To evaluate how compressed bed height influences chromatography efficiency.”

CASE 1 — Compressed Bed Height: 18.2 cm

Column Efficiency Evaluation - Studied at linear flow velocities of 50, 100, 150 & 200 cm/hr



Integration Result:UV1_PeakTable

Number	Peak	Retention[mL]	Area	Height	Plates per meter	Plate height	Asymmetry	Separation	Trailing factor
1	50 cm/Hr	407.91	456.98	7.8	1277.28	0.07829	1.3	0	0.75
2	100 cm/Hr	1056.313	554.16	7.78	7158.97	0.01397	1.19	10.83	0.78
3	150 cm/Hr	1621.049	570.02	6.44	11934.79	0.00838	1.09	7.8	0.64
4	200 cm/Hr	2194.275	618.36	5.41	14897.81	0.00671	1.16	6.5	0.76

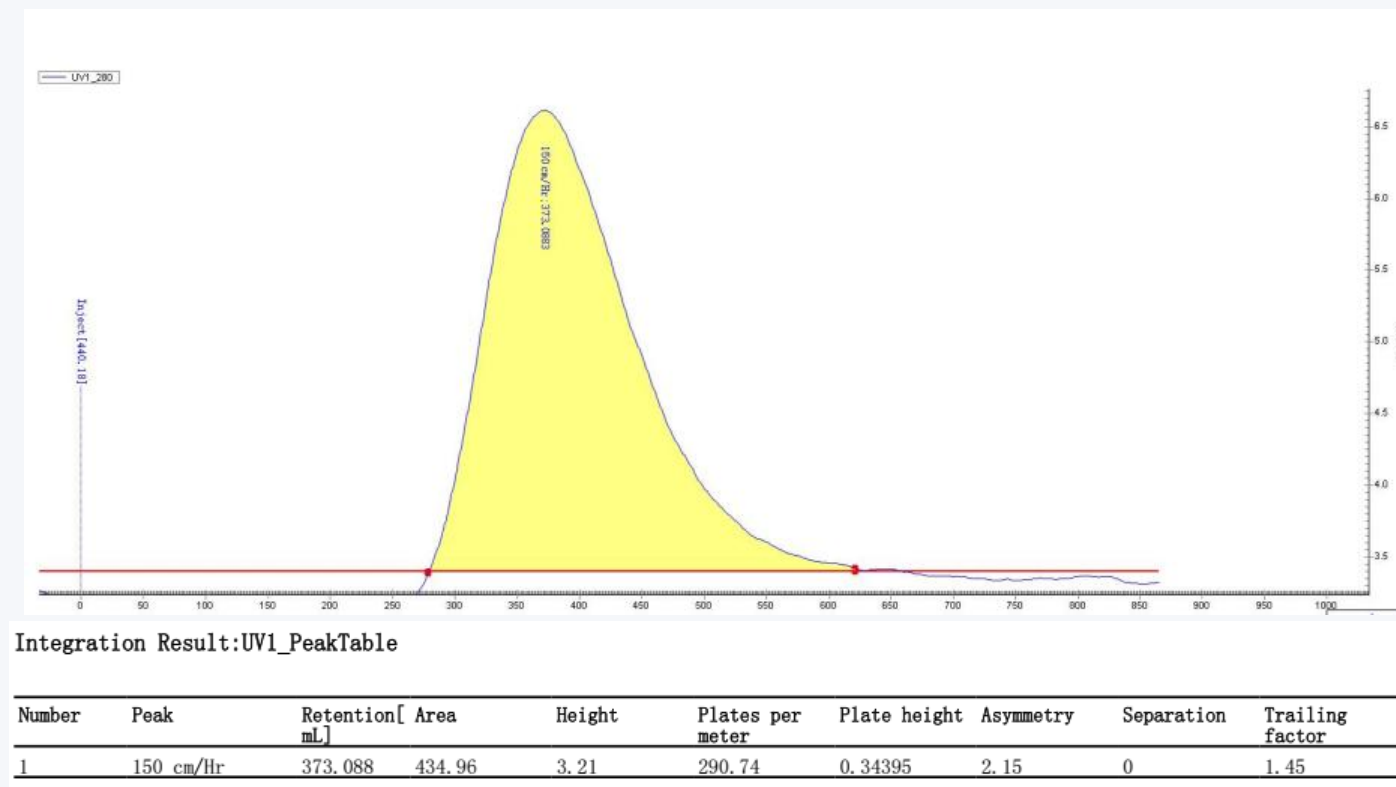
Key Observations:

- ✓ Improved peak symmetry
- ✓ Better asymmetry values
- ✓ Higher plates per meter
- ✓ Lower plate height
- ✓ More consistent performance across flow velocities

“Improved bed consolidation resulted in better peak symmetry and higher column efficiency.”

CASE 2 — Compressed Bed Height: 17 cm

Column Efficiency Evaluation - Studied at linear flow velocities # 150 cm/hr



Observations

- ✓ Broader peak profile
- ✓ Higher asymmetry
- ✓ Lower plates per meter
- ✓ Increased plate height
- ✓ Reduced overall column efficiency

Practical Observation: Even a relatively small difference in bed consolidation influenced chromatography performance significantly.

Case 1 (18.2 cm)

- ✓ Sharper peak
- ✓ Better asymmetry
- ✓ Higher Plates/meter

Case 2 (17 cm)

- ⚠ Broader peak
- ⚠ Higher tailing
- ⚠ Lower Plates/meter

Key Learning

Column packing quality can strongly influence chromatography reproducibility — even when using the same resin, column and packing conditions.

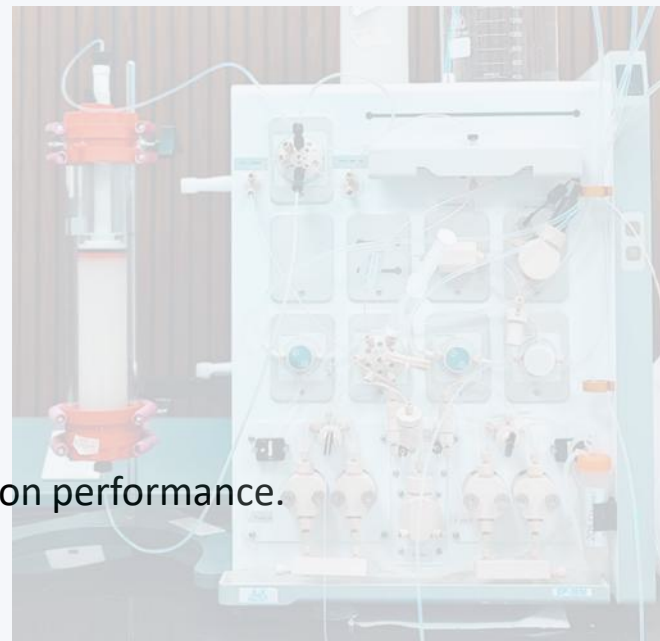
In downstream purification workflows,

Reproducibility often begins during column packing itself.

Careful optimization of:

- Packing protocol
- Bed consolidation
- Flow conditions
- Column efficiency evaluation

can significantly improve process consistency and purification performance.



Parameters such as: ✓ HETP ✓ Asymmetry ✓ Plate height ✓ Plates per meter
are valuable tools for evaluating packed bed quality.

“Do you routinely evaluate HETP and asymmetry during column packing studies?”

ProtPure Tech Pvt. Ltd.

Building chromatography science and purification solutions in India.